

QNH Budget Management System

Executive Overview

The QNH Budget Management System creates one controlled process connecting:

Financial Year

→ Department Planning

→ Budget Submission

→ Approval or Return

→ Transfers and PO Linking

→ Consumption Tracking

→ Analytics and Reporting

→ Financial Year Closing

- **Business purpose:** replace fragmented budget preparation, review, procurement tracking, and audit evidence with one governed workflow.
- **Main financial controls:** financial year status, budget approval, return feedback, transfer approval, PO link approval, permissions, and audit logs.
- **Primary users:** Department Heads, Approvers / CFO, Finance, PO Approval Team, Budget Administrators, and System Administrators.
- **Executive value:** clearer budget status, stronger approval discipline, procurement-to-budget visibility, and better evidence for price decisions.
- **Demonstration focus:** how a department budget moves from planning to approval, then into controlled transfers, PO consumption, analytics, and audit visibility.

1. Why the System Exists

Hospital budgeting requires more than entering numbers. Finance leadership needs a controlled process showing who requested what, who approved it, how spending consumes the approved budget, and whether requested prices are reasonable.

Before	With the QNH Budget System
Manual and fragmented	One controlled workflow
Limited approval visibility	Clear statuses and action queues
Informal return feedback	Documented general and item-level notes
Uncontrolled balance movement	Approved transfer workflow
PO activity disconnected	PO activity linked to approved budget
Manual price investigation	Historical price benchmarks

Before**With the QNH Budget System**

Scattered evidence

Central audit trail

The system reduces dependency on spreadsheets, email follow-up, and personal knowledge. It gives Finance a consistent way to govern planning, approvals, procurement linkage, and year-end closure.

2. Who Uses the System

The system has four main user roles. Each role has a clear responsibility within the budget lifecycle.

Role	Main Responsibility	Main Actions and Decisions
HOD	Prepare and manage the department budget	Decide what the department needs, submit the budget, and correct returned items
Approver / CFO	Control financial years and approve department budgets	Open and close financial years, approve budgets, or return them for correction
Finance / PO Approval Team	Control PO linking and budget consumption	Approve valid PO links and monitor consumed and remaining balances
Administrator	Manage system setup and user access	Manage users, permissions, budget items, categories, and PO mappings

Department Head — HOD

The HOD prepares and manages the department budget.

Main activities

- Create and edit budget items.
- Enter the quantity and unit price.
- Select the item distribution method.
- Save the budget as a draft.
- Copy items from an approved historical budget.
- Import multiple budget items from Excel.
- Request a new item type when the required item does not exist.
- Mark important budget items as project items.
- Submit the completed budget for approval.
- Review general and item-level return notes.
- Correct and resubmit returned budgets.
- Request budget transfers where permitted.
- Monitor the department budget status, consumption, and remaining balance.

Main responsibility

The HOD decides **what the department needs** and provides the required budget information.

Approver / CFO

The Approver or CFO controls the financial-year lifecycle and reviews submitted department budgets.

Financial-year activities

- Create a new financial year.
- Open the financial year for department budget preparation.
- Monitor the progress of department budget submissions and approvals.
- Move the financial year from **OPEN** to **PRE-CLOSING** after all department budgets are approved.
- Confirm that pending transfers, PO links, and required activities are resolved.
- Move the financial year from **PRE-CLOSING** to **CLOSED** when all closing conditions are complete.

Budget-approval activities

- Review department budget totals.
- Review individual budget items.
- Review quantities, unit prices, and total amounts.
- Clearly distinguish the unit price from the total item amount.
- Review item distribution across the year.
- Review project-item classifications.
- Use Price Intelligence to compare requested prices with historical purchasing data.
- Review benchmark variance and potential overspend.
- Use dashboards and analytics to compare departments and identify risks.
- Approve acceptable budgets.
- Return budgets that require correction, clarification, or justification.
- Add general feedback or notes for specific items.

Main responsibility

The Approver or CFO decides:

- **Whether a submitted department budget is acceptable.**
 - **When all department budgets have completed the approval process.**
 - **When the financial year should move to PRE-CLOSING.**
 - **When the financial year is ready to be CLOSED.**
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Finance Department / PO Approval Team

Finance and the PO Approval Team control how approved purchase orders consume approved budgets.

Main activities

- Review submitted PO link requests.
- Confirm that the PO is linked to the correct approved budget item.
- Check the requested PO quantity.
- Check the remaining available PO quantity.
- Check the available quantity and balance of the budget item.
- Approve valid PO links.

- Reject invalid PO links and provide a reason.
- Monitor approved, consumed, transferred, and remaining budget amounts.
- Review transfer activity where assigned.
- Resolve pending PO links and financial activities during PRE-CLOSING.
- Support the CFO in confirming financial-year closure readiness.
- Use dashboards, analytics, and reports for financial monitoring.

Main responsibility

Finance decides **whether a PO link is valid and should consume the selected approved budget item.**

Finance supports the closing process, but the decision to move the financial year between **OPEN, PRE-CLOSING**, and **CLOSED** belongs to the Approver / CFO.

Administrator

The Administrator manages the system setup and user access required for the budget process.

Main activities

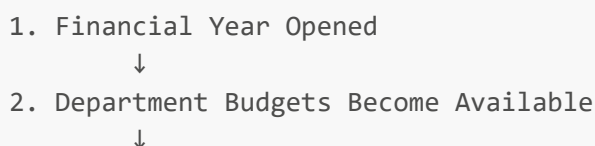
- Manage users and department access.
- Assign roles and permissions.
- Control who can view, edit, submit, approve, or administer each area.
- Manage budget categories.
- Manage budget item types.
- Review and approve or reject new budget-item requests.
- Maintain mappings between budget item types and PO item codes.
- Activate or deactivate categories, item types, and mappings where permitted.
- Review audit and access information where authorized.
- Maintain configuration required for budget entry, approval, PO linking, analytics, and reporting.
- Support users with access or configuration issues.

Main responsibility

The Administrator decides:

- **Which users can access the system.**
- **Which roles and permissions each user receives.**
- **Which budget categories and item types are available.**
- **Which PO Item Mappings are active.**

3. End-to-End System Lifecycle



3. HOD Creates, Copies, or Imports Budget Items
↓
4. HOD Reviews and Submits the Department Budget
↓
5. CFO / Budget Approver Reviews the Submitted Budget
↓
6. Budget Is Approved or Returned for Correction
↓
7. Returned Budget Is Corrected, Resubmitted, and Reviewed Again
↓
8. All Department Budgets Are Approved
↓
9. Financial Year Moves to PRE-CLOSING
↓
10. Approved Budgets Can Receive Transfer Requests and PO Link Requests
↓
11. Transfers and PO Links Are Reviewed and Approved or Rejected
↓
12. Approved Transfers Update Budget Item Balances
↓
13. Approved PO Links Update Budget Consumption and Remaining Balances
↓
14. Dashboards, Analytics, Reports, Emails, and Audit Logs Monitor Activity
↓
15. All Pending Transfers, PO Links, and Required Activities Are Resolved
↓
16. Financial Year Is CLOSED and Preserved for Historical Reporting

Example:

Radiology opens its current budget, adds ultrasound equipment and related items, imports other planned items from Excel, and submits the budget. The CFO reviews the item prices and Price Intelligence indicators. If a price needs explanation, the budget is returned with notes. After correction and approval, future PO links consume the approved budget, and Finance can monitor remaining balance and closure readiness.

4. Financial Year Lifecycle

Status	Main Purpose	What Happens
OPEN	Active planning and approval	Budgets are created, submitted, and reviewed
PRE-CLOSING	Resolve remaining activity	Pending transfers and PO links are addressed
CLOSED	Preserve final history	New active changes are restricted

Why this matters to the CFO:

- Creates a clear annual control period.
- Prevents uncontrolled activity after closure.
- Requires unresolved transfer and PO-link activity to be addressed.

- Supports historical reporting and audit.

5. Department Budget Preparation

Normal HOD workflow:

```
Open Current Budget
→ Add or Reuse Items
→ Enter Quantity and Unit Price
→ Select Distribution
→ Save Draft
→ Review Total
→ Submit
```

Manual Budget Entry

Each budget item includes the main financial planning details:

Field	Meaning
Category	Budget grouping.
Item type	Specific requested item or service.
Quantity	Number of units requested.
Unit price	Expected cost per unit.
Total amount	Quantity multiplied by unit price.
Distribution method	How quantity is planned during the year.
Project item	Marks the item for elevated visibility and reporting.

Distribution Methods

Method	Meaning
Annual	One annual quantity.
Monthly	Quantity distributed across months.
Quarterly	Quantity distributed across quarters.
Custom	User-defined monthly or quarterly distribution.

Control:

Total distributed quantity must equal the requested item quantity.

Copy Budget From History

Users can copy items from an approved historical budget and adjust them for the new financial year.

Business value:

- Faster annual planning.
- Less repetitive entry.
- Reuse of previously approved items.

Excel Import

Download Template
→ Complete Excel File
→ Import
→ Review Valid Rows and Errors
→ Correct if Needed

Business value:

- Faster entry for large budgets.
- Standardized template.
- Clear error feedback.

Budget Item Requests

Required Item Does Not Exist
→ HOD Submits Item Request
→ Administrator Reviews
→ Approves or Rejects
→ Approved Item Becomes Available

This allows departments to request new item types without bypassing administrator control.

Project Items

Project = Budget item marked for elevated visibility and reporting.

This is not currently a full project-management platform. It is a budget-driven way to highlight important budget items.

6. Budget Submission, Approval, and Return

Status	Meaning
DRAFT	Editable by the department
PENDING_APPROVAL	Submitted and locked
RETURNED	Editable again for correction
APPROVED	Accepted as the approved budget plan

Workflow:

```
HOD Submits  
→ Budget Is Locked  
→ Approver Reviews  
→ Approve or Return
```

The Approver reviews:

- Item.
- Quantity.
- Unit price.
- Total amount.
- Distribution.
- Project classification.
- Benchmark and variance.
- Potential overspend.
- Supporting notes.

Return loop:

```
Approver Returns Budget  
→ General and/or Item Notes Are Recorded  
→ HOD Corrects Budget  
→ HOD Resubmits  
→ Approver Reviews Again
```

Example:

If Office Chair is requested at SAR 420 while historical benchmark is SAR 390, the CFO may return the item asking the department to confirm specification or revise the unit price.

7. Budget Transfers

Approved budget value can be moved in a controlled way when business priorities change.

Source Budget Item
 → Transfer Request
 → Approval or Rejection
 → Destination Budget Item
 → Balances Updated

Item	Before	Change	After
Furniture	SAR 100,000	- SAR 20,000	SAR 80,000
Medical Equipment	SAR 300,000	+ SAR 20,000	SAR 320,000

Controls:

- Valid source item.
- Valid destination item, or approved creation of a new budget item.
- Available transferable balance.
- Approval required.
- Full audit history.

Verified capability:

The transfer process supports existing-item transfers and new-item transfers. When an approved transfer is for a new item, the approved process creates the new budget item and allocates the requested amount.

8. Purchase Order Linking and Approval

PO Linking connects actual purchasing activity to the approved budget item it consumes.

PO Record
 → User Selects Approved Budget Item
 → PO Link Request
 → Finance / PO Approval Review
 → Approve or Reject
 → Approved Consumption Updated

Measure	Amount
Approved Budget	SAR 100,000
Approved PO Link	SAR 30,000
Consumed	SAR 30,000
Remaining	SAR 70,000

Key controls:

- Budget item must be eligible and approved.
- Requested quantity is validated.
- PO available quantity is validated.
- Budget item available quantity is validated.
- Approval is required.
- Rejection reason can be captured.
- Pending requests are considered before approval and closing.

This gives Finance a clear connection between procurement activity and approved budget consumption.

9. PO Item Mapping

PO Item Mapping tells the system which procurement item codes belong to each budget item type.

```
Budget Item Type
→ One or More Mapped PO Item Codes
→ Historical Purchase Records
→ PO Suggestions and Price Intelligence
```

Example:

```
Budget Item Type: Laptop
Mapped PO Item Code: 17819
Result: historical purchases for item 17819 can support Laptop analysis.
```

Mapping sources:

- **Manual mapping:** maintained by an administrator.
- **Approved-link mapping:** learned from an approved PO link.

Why mappings matter:

- Better PO suggestions.
 - Better historical comparisons.
 - More reliable benchmark evidence.
 - Reduced dependence on free-text descriptions.
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10. Budget Consumption and Remaining Balance

Value	Meaning
Approved Amount	Budget value approved for the item
PO Consumed	Value consumed by approved PO links
Transfer In	Approved value added
Transfer Out	Approved value removed
Remaining	Value still available

Formula:

```
Remaining Budget
=
Approved Budget
- Approved PO Consumption
+ Approved Transfer In
- Approved Transfer Out
```

Example:

Measure	Amount
Approved Budget	SAR 250,000
Approved PO Consumption	SAR 90,000
Approved Transfer In	SAR 20,000
Approved Transfer Out	SAR 0
Remaining Budget	SAR 180,000

This is the financial bridge between planning and actual budget usage.

11. Price Intelligence

Price Intelligence does not automatically approve or reject an item. It gives the approver historical procurement evidence for a better-informed decision.

Data flow:

```
Budget Item
→ PO Item Mappings
→ Historical PO Records
→ Benchmark Calculations
→ Approval Indicators
```

Metric	Meaning	Simple Calculation
Historical Benchmark	Typical historical unit price	Median historical unit cost
Variance Amount	Unit-price difference	Budget Unit Price - Benchmark
Variance %	Difference as a percentage	Variance / Benchmark x 100
Potential Impact	Total difference for quantity	Variance x Quantity
Potential Overspend	Positive risk amount only	$\max(0, \text{Variance}) \times \text{Quantity}$
Evidence Strength	Reliability of evidence	Purchase count, suppliers, and recency

Complete example:

Requested Quantity: 50
 Budget Unit Price: SAR 420
 Historical Benchmark: SAR 390

Variance Amount:
 $\text{SAR } 420 - \text{SAR } 390 = \text{SAR } 30$

Variance Percentage:
 $\text{SAR } 30 / \text{SAR } 390 \times 100 = 7.7\%$

Potential Overspend:
 $\text{SAR } 30 \times 50 = \text{SAR } 1,500$

Why median is used:

Historical Prices:
 90, 95, 100, 105, 500

Average: 178
 Median Benchmark: 100

The median is less distorted by the unusual SAR 500 purchase.

Price Intelligence statuses:

- Within Benchmark.
- Review Price.
- Significant Variance.
- High Overspend Risk.
- No Benchmark Available.

Approver use:

The CFO should treat these indicators as decision support. A high variance does not automatically reject an item, but it gives a clear reason to ask for justification, revised pricing, or specification clarification.

12. Dashboards, Analytics, and Reports

Capability	Main Purpose
Dashboard	Shows current status and actions requiring attention
Analytics	Lets users filter, compare, and investigate
Reports	Provides formal operational and management outputs

Dashboards

Dashboards provide role-based visibility:

- HOD view: budget status, returned items, department actions.
- Approver / CFO view: pending approvals, workload, financial exposure, and high-risk items.
- Operational panels: transfers, PO links, item requests, and other workflow actions.

Budget Analytics

Analytics helps answer:

- Which departments have the largest budgets?
- Which budgets are approved, returned, or pending?
- Which items have high variance?
- Which items have missing benchmarks?
- Which categories drive the largest values?

Reports

Current state:

- The dedicated **Reports module is not implemented yet.**
- Advanced reporting remains a planned feature.
- The Approver / CFO can export the currently displayed data or filtered view to Excel from supported system pages.
- Excel export can be used for additional review, analysis, sharing, or offline records.
- Planned report categories include budget reports, approval reports, transfer reports, PO linking reports, consumption reports, Price Intelligence reports, and audit reports.

The current Excel export capability should not be treated as a complete Reports module.

Planned reports should not be presented as fully operational until the dedicated Reports module is implemented.

13. Email Notifications

Trigger	Recipient	Purpose
Budget submitted	Approver	Review required
Budget approved	HOD / owner	Approval confirmed
Budget returned	HOD / owner	Correction required
Transfer submitted	Transfer approver	Review required
Transfer approved or rejected	Requester	Outcome communicated
PO link submitted	PO approver	Review required
PO link approved or rejected	Requester	Outcome communicated
Item request created	Administrator	Setup review required
Item request approved or rejected	Requester	Outcome communicated
Financial year opened, pre-closed, or closed	Relevant users	Lifecycle status communicated

Notifications reduce manual follow-up and keep users aware of required actions.

14. Permissions and Segregation of Duties

Users only see and perform actions appropriate to their assigned responsibilities.

Action	HOD	Approver / CFO	Finance / PO Approver	Admin
Create department budget	Yes	According to access	According to access	According to access
Submit budget	Yes	According to access	According to access	According to access
Approve or return budget	No	Yes	Only if assigned	Only if assigned
Request transfer	If permitted	If permitted	If permitted	If permitted
Approve transfer	No unless assigned	If permitted	If permitted	If permitted
Approve PO link	No unless assigned	If permitted	Yes if assigned	If permitted
Manage configuration	No	No	Limited	Yes if authorized
Manage user access	No	No	No	Yes

CFO value:

- Segregation of duties.
- Controlled authority.
- Restricted visibility.
- Reduced unauthorized changes.
- Clear ownership of approvals and administration.

15. Audit Logs and Governance

Audit logs help answer:

- Who performed the action?
- When did it happen?
- What changed?
- What was the decision?
- Why was an item returned or rejected?

Tracked areas include:

- Financial years.
- Budget submission, approval, and return.
- Transfers.
- PO links.
- Item requests.
- Access or configuration actions where implemented.

Governance model:

Permissions

→ Authorized Actions

→ Approval Workflows

→ Email Notifications

→ Audit Logs

→ Management Visibility

Together, these features improve accountability and reduce reliance on informal records.

16. Current Capabilities and Roadmap

Capability	Status
Financial Years	Implemented
Budget Entry and Submission	Implemented

Capability	Status
Copy From History	Implemented
Excel Import	Implemented
Item Requests	Implemented
Approval and Return Feedback	Implemented
Budget Transfers	Implemented
Transfers to New Budget Items	Implemented
PO Linking and Approval	Implemented
PO Item Mapping	Implemented
Consumption Tracking	Implemented
Price Intelligence	Implemented
Dashboards and Analytics	Implemented
Notifications	Implemented
Permissions and Audit Logs	Implemented
Project-item foundation	Implemented / Partial
Advanced Reporting	Partial / Planned
Companies and Vendors	Planned
Smart Analytics Filters	Planned

Roadmap items are not presented as current operational functionality.

17. Executive Conclusion

- One controlled budget lifecycle from financial year opening to closure.
- Clear responsibility for every major role.
- Stronger approval, return, transfer, and PO-link controls.
- Procurement activity connected to budget consumption and remaining balance.
- Better decisions through historical evidence, analytics, notifications, permissions, and auditability.